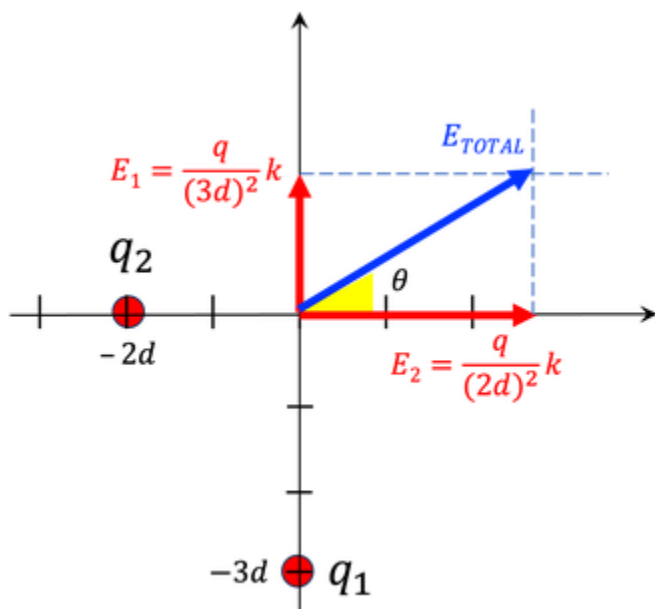


PHY2049 Final December 2023

1. The Electric field is proportional to the charge and inversely proportional to the distance squared. So, therefore, at the origin, the electric field has an x-coordinate of $+kq/4$ and a y-coordinate of $kq/9$. The tan of the angle with respect to the x-axis is therefore $4/9$, and this gives 24° .



$$\tan \theta = \frac{E_1}{E_2} = \frac{4}{9}$$

$$\theta = 24^\circ$$

2. After connection, the two have the same voltage difference. Therefore, as the charges are in proportion to the capacitance ($Q = CV$) and charge is conserved, each now have a potential of 2V. Before, energy = $0.5CV^2 = 0.5 \times 1 \times 10^{-6} \times (10^4)^2 = 50 \text{ J}$
After, we have $0.5 \times 5 \times (2 \times 10^3)^2 = 10 \text{ J}$
A difference of 40 J
3. As you stretch a wire its length doubles, but its cross section decreases by a factor of 2. The result is that the resistance increases by a factor of 4. As the heat dissipated by a resistor in a simple circuit is V^2/R , this means that the heat dissipated will decrease by a factor of 4.
4. As shown in Exam 1, Q9, the electric field inside a slab, a distance d_2 from one surface and d_1 from another, is given by: $E = \rho(d_2 - d_1)/2\epsilon_0$

Therefore $(d_2 - d_1) = 34 \times 2 \times 8.85 \times 10^{-12} / (10 \times 10^{-9}) \text{ m}$
 $= 0.06 \text{ m}$

That is the 12 cm from one side and 18 cm from the other.

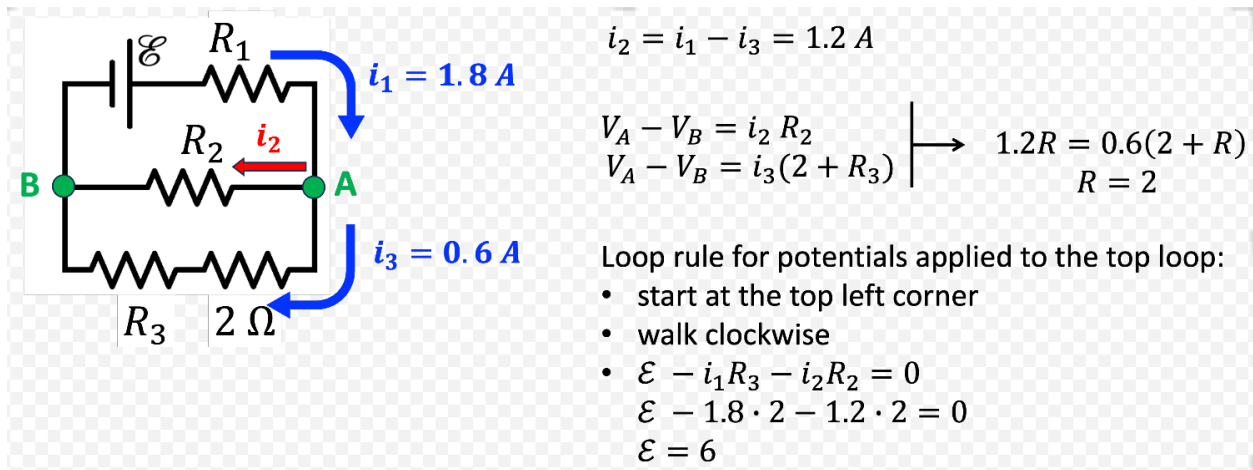
5. The particle starts with kinetic energy K and comes to stop. The force on it is given by:

$$qE = q\sigma/\epsilon_0$$

Energy conservation gives $0.5mv^2 = Dq\sigma/\epsilon_0$

Therefore $v^2 = 2Dq\sigma/m\epsilon_0$

6. See diagram.



7. Taking up to be positive, and knowing that the force being proportional to $1/r$ and is attractive for currents going in the same direction as each other, we find the force to begin with on wire B to be :

$$(\text{constant}) \times (-1/d - 1/d + 1/2d) = \text{constant} \times (3/2d)$$

where the three terms are due to wires A, C and D and d is the wire spacing.

Now reverse the last term as we reverse D's direction. We get a new answer of:

$$(\text{constant}) \times (5/2d).$$

The ratio of the two is $5/3$.

8. Ampere's Law tells us that $\mathbf{B} \cdot d\mathbf{s} = B \times 2\pi(0.001) = 4\pi \times 10^{-7} \times 0.2 \times (0.001/0.005)^2$

$$B = 2 \times 10^{-4} \times 0.2 \times 1/25 = 1.6 \times 10^{-6} \text{ T}$$

(Note that the $(0.001/0.005)^2$ is the fraction of the current that is closer to the center than the point of interest).

9. We know that $d\Phi_B/dt = A dB/dt = \varepsilon$

$$dB/dt = 0.4 - 0.2t \text{ T/s}$$

$$\text{At } t = 4 \text{ s, } dB/dt = -0.4 \text{ T/s}$$

$$A = 0.02 \times 0.02 \text{ m, so we get } B = 1.6 \times 10^{-6} \text{ T}$$

(The sign does not matter as we are asked for the magnitude).

10. At resonance, the capacitive and inductive reactances effectively cancel (i.e. are equal), leaving $R = Z$

(The voltage amplitude over the inductor may be big or small - we don't know, we don't know if the reactances are big or small, and the voltages across the capacitor and inductor are 180° out of phase).

Unfortunately, by some quirk of editing, there were 6 answers offered, which made it impossible for some people to answer correctly on the scantron. Therefore, we marked everyone correct for this question.

11. The magnetic field along a closed loop (by symmetry a circle in this case), is $B2\pi r$ and is proportional to the rate of change of flux. That flux is proportional to area and thus to r^2 . Thus B is proportional to r^2/r .

12. A superconductor is an extreme case of diamagnetism - that is how it was described in class, and that is the answer we wanted. Clearly this is not any of the other three types of magnetism listed. However, we were sympathetic to those who interpreted "no magnetism" as meaning "no B field inside it", so we allowed that as an alternative.

13. The entire power of 1MW is transmitted to the asteroid, so the intensity is $I = 1 \text{ MW}/A$, where A is the area, and the pressure is $1\text{MW}/(cA)$ (intensity divided by c , as shown on formula sheet). As force is (pressure $\times A$), we have A conveniently canceling.

14. The intensity of unpolarized light getting through the first is 0.5 (as is always the case for unpolarized light going through a polarizer). Then this light is polarized at an angle 30° to the second, so that gives a factor of $\cos^2(30^\circ)$ at the second polarizer, then the transmitted light from the second is polarized at an angle of 30° to the polarization plane of the third which gives a further $\cos^2(30^\circ)$. That makes $0.5\cos^4(30^\circ) = 0.28$ (i.e. 28%)

15. Plane and convex mirrors always produce a virtual image. Concave mirror can in some circumstances.

16. The image for a very distant object is very close to the focal point (parallel rays all meet at the focal point). So, the image is at approximately 5 mm from the lens. The magnification is proportional to ratio of the image and object distances, i.e.

$m = 5 \times 10^{-3} / 3.84 \times 10^8$. So, we multiply the object size to get (approximately)

$$(5 \times 3.6 / 3.85 = 5) \times 10^{-3-8+6} = 5 \times 10^{-5} \text{ m} = 0.05 \text{ mm}$$

17. You cannot use the standard formula here, because the angles are not small and so the approximations do not work. Simply work out the differences in the distance and set that to be half a wavelength because half a wavelength difference will indicate the first minimum. For the closer antenna, the distance is $d_1^2 = 200^2 + 200^2 = 281 \text{ m}$, for the second one it is $d_2^2 = 200^2 + 400^2 = 447$

So, wavelength = $2 \times (447 - 281) = 332 \text{ m}$

18. We want maximized intensity, so the reflections between the surfaces are in phase. Off the front face, there is a phase change of π radians (half a wavelength) as we go from a less dense to a more dense medium. The ray that goes to the back surface has a further $2d$ with a wavelength in that medium of $\lambda = 450 / 1.5 = 300 \text{ nm}$. So, the extra phase is $4\pi d / 300$ where d is in nm.

So, we have $2\pi d / 300 - \pi = 0, 2\pi, 4\pi$ etc.

$d = 75, 225, 375, 725$ etc., so 225 nm is the one that fits.

19. The formula for the diffraction grating maxima of order "m" is $2d \sin \theta = m\lambda$ and thus d is proportional to the sine of the angle. Note that we are not in the realm of small angles, so we cannot put $\sin \theta = \theta$ and the answer is $\sin^{-1}(2 \times \sin 20^\circ) = 43^\circ$

20. The first minimum is given by $a \sin \theta = \lambda$ and we are in the realm of very small angles, so $\sin \theta = \theta = y / D$ where y is 1 cm. So,

$$1 \times 10^{-4} \times 10^{-2} / D = 5 \times 10^{-7}$$

$D = 2 \text{ meters}$.